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10 August 1982
SUPERSEDING
MIL-H-46709C(AT)
6 December 1976

MILITARY SPECIFICATION

HOWITZER, LIGHT, SELF-PROPELLED: 105MM, M108; AND HOWITZER, MEDIUM, SELF-PROPELLED: 155MM, M109, M109A1, M109A1B, M109A2; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the processing of the Howitzer, Light, Self-Propelled: 105MM, M108; and Howitzer, Medium, Self-Propelled: 155MM, M109, M109A1, M109A1B, M109A2 for domestic or oversea shipment, including carloading meeting the requirements of levels A and B processing (see 1.2).

1.2 Classification. Processing shall be of the following levels as specified (see 6.1).

Level A - Maximum Military Protection. Processing for domestic or oversea shipment, including open deck loading and any outside storage in excess of 90 days from date of actual processing at the production facility (periodic care and preservation during storage required).

Level B - Minimum Military Protection. Limited processing for immediate use shipment and for domestic or oversea shipment (excluding open deck grating) and any storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

No deliverable data required by this document.

AREA-PACK

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not to exceed 90 days from date of actual processing at production facility. Provision is made for drive-away capability when specified (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Twin Gage, (Polyolefin).
O-A-548	- Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated.
O-E-760	- Ethyl Alcohol (Ethanol); Denatured alcohol; Proprietary Solvent and Special Industrial Solvents.
O-S-576	- Sodium Bicarbonate, Technical.
P-D-220	- Detergent, General Purpose.
GG-T-250	- Tester, Antifreeze Solutions.
QQ-S-781	- Steel Strapping and Seals.
QQ-A-1876	- Aluminum Foil.
RR-W-360	- Wire Fabric, Industrial.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
UU-P-268	- Paper, Kraft, Wrapping.
UU-T-81	- Tag, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
VV-L-800	- Lubricating Oil, General Purpose, Preservative, (Water-Displacing, Low Temperature).
ZZ-T-416	- Tire, Pneumatic: Retread and Repair Materials.
MMM-A-178	- Adhesive, Paper Label, Water-Resistant.
NNN-P-40	- Paper, Lens.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-T-42	- Tape, Packaging/Masking, Paper.
PPP-T-60	- Tape: Packaging Waterproof.
PPP-T-97	- Tape: Pressure-Sensitive Adhesive, Filament Reinforced.

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bags, Sleeve and Tubing, Interior Packaging.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-S-207	- Sulfuric Acid, Electrolyte: Packaging, Packing, and Marking for Shipment and Storage of.
MIL-C-450	- Coating compound, bituminous solvent type, Black (for Ammunition).
MIL-P-3420	- Packaging Materials, Volatile Corrosion Inhibitor, Treated, Opaque.
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-H-6083	- Hydraulic fluid, petroleum base, for Preservation and Operation.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-B-11188	- Battery, Storage, Lead-Acid.
MIL-B-12841	- Basic Issue Items for Military Vehicles, Carriages and Equipment; Preparation for Shipment and Storage of.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-C-16173	- Corrosion Preventitive Compound, Solvent Cutback, Cold-Application.
MIL-D-16791	- Detergents, General Purpose (Liquid, Nonionic).
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-in.
MIL-B-22191	- Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-L-46002	- Lubricating Oil, Contact and Volatile Corrosion Inhibited.
MIL-D-81298	- Dye, Liquid, for Detection of Leaks in Aircraft Fuel Systems.

2.1.2 Other Government documents, and Government drawings and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

10941519	- Vehicle Closure Kit M108, M109, M109A1
12260206	- Vehicle Closure Kit M109A1B, M109A2
12260160	- Installation OVE Saddle Depot OVE Box
12260223	- Installation Ventilation Kit.

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PUBLICATIONS

DEPARTMENT OF TRANSPORTATION

Hazardous Materials Regulations

(Application for copies should be addressed to the Department of Transportation, Hazardous Materials Regulations Board, Washington, D.C. 20590.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASSOCIATION OF AMERICAN RAILROADS

Section No. 1	- General Rules Governing Loading of Commodities on Open Top Cars.
Section No. 6	- Rules Governing the Loading of Department of Defense Materiel on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, Illinois 60605.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Level A.

3.1.1 First production processed vehicle. Unless otherwise specified (see 6.2), the first production processed vehicle shall be subjected to the inspection specified in 4.2. Approval of the first production processed vehicle shall not relieve the contractor of his

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obligation to process all vehicles in accordance with this specification. Except as otherwise specified by the procuring activity, any change to materials or design after approval of the first production processed vehicle shall require submission of an additional production processed vehicle to inspection specified in 4.2.

3.1.2 One of the next 10 production processed vehicles. After approval of the first production processed vehicle (see 3.1.1), one of the next 10 production processed vehicles shall be selected by the Government representative for inspection specified in 4.2.

3.1.3 Stock numbers. Materials to be obtained through Government supply channels shall be ordered by stock numbers or specifications.

3.1.4 Preparation prior to processing. Except as otherwise specified herein, and to the maximum extent consistent with production efficiency, economy, and safe storage and shipment, vehicle shall be prepared for storage and shipment in a completely assembled condition after test runs, and completion and approval of all necessary repairs. Specified equipment shall be installed and all adjustments made so that vehicle may be operated, shipped, and placed into service with a minimum of delay.

3.1.4.1 Processing records. Records of vehicle processing shall be maintained and readily available for review by the Government representative.

3.1.4.2 Disassembly. Projecting parts whose removal will accomplish desired cube reduction and parts susceptible to damage and pilferage shall be removed from the vehicle. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured. Packed parts shall be identified and stowed securely within the vehicle.

3.1.4.2.1 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be put on cloth shipping tags conforming to type A of UU-T-81, or on metal tags marked with soluble paint, and attached to mating parts. The marked cloth shipping tags shall be waterproofed with varnish conforming to MMM-A-178.

3.1.5 Cleaning and drying.

3.1.5.1 Interior of vehicle. Interior surfaces of vehicle shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791 and warm water. Water or other liquid under pressure, or steam cleaning shall not be used. After cleaning, cleaned surfaces

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shall be rinsed with clean water and dried. Care shall be taken during cleaning and rinsing to assure that no liquids enter instruments, connections, or other components susceptible to water damage and that water does not accumulate in areas where it cannot drain or be dried.

3.1.5.1.1 Battery supports and retainers. Battery supports and retainers shall be cleaned with a solution of one-half pound of sodium bicarbonate conforming to O-S-576 per gallon of water. After cleaning, cleaned surfaces shall be flushed with clean water, then thoroughly dried. Dried surfaces shall then be preserved in accordance with 3.1.12.1.

3.1.5.1.2 Backrests and seats. Backrest and seat cushions shall be cleaned with a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, in warm water. After cleaning, cushions shall be wiped with cloths saturated with clean water to remove cleaning solution. Care shall be taken not to saturate the cushions with cleaning solution or water. After rinsing, cushions shall be dried, then protected in accordance with 3.1.16.

3.1.5.1.3 Fire control items. Exposed optical glass components of fire control items shall be cleaned by blowing exposed optical glass surfaces with air from a hand syringe, or by use of clean camel-hair brush, followed by the use of ethyl alchohol conforming to O-E-760. In cases of contamination not removed by alcohol, cleaning shall be accomplished by use of a solution consisting of two ounces of detergent conforming to type I of MIL-D-16791, 1/2 gallon of alcohol conforming to O-E-760, and one gallon distilled water. Using a swab made of paper conforming to type I or II of NNN-P-40, optical glass surfaces shall be washed with the cleaning agent; and washing shall be repeated, using a clean swab each time, until no dirt or other foreign matter remains on the surface. Cleaning shall be accomplished with a minimum of pressure and rubbing, and without use of cloth or rubber materials. Immediately after cleaning, the optics shall be covered or wrapped with lens tissue conforming to NNN-P-40, and secured with tape conforming to PPP-T-42.

3.1.5.2 Exterior of vehicle. Using a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, and water or steam, vehicle exterior surfaces shall be cleaned in a manner which will assure removal of all foreign matter. After cleaning, surfaces cleaned shall be rinsed with water and thoroughly dried. Exterior of vehicle shall be cleaned without permitting entry of steam or water into driver's or engine compartments.

3.1.5.2.1 Cannon. When inspection indicates the need for reprocessing (see 4.5.2.3), cannon shall be cleaned in accordance with process C-3 and thoroughly dried in accordance with procedure D-4 of MIL-P-116.

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3.1.6 Relubrication. When vehicle has been operated in excess of 50 miles since lubrication, or after cleaning with detergent or steam cleaning, vehicles shall be lubricated using materials in accordance with drawings, specifications or lubrication order pertinent to the vehicle. All exposed oil can points such as levers, driver's periscope cover rotating surfaces, periscope cover spring-loaded trunnions, locking bars, strikers, hinges, hinge pins, locking pins, pintle pins, locking levers, wing nuts, latches, door locks, hand-operated locking knobs, linkage, and threaded ends of yokes and related clevis pins shall be lubricated with oil conforming to VV-L-800. Excess lubricating material shall be removed after lubrication.

3.1.7 Transmission, transfer assembly, control differential, and final drives. Transmission shall contain lubricating oil only, conforming to Type I, Grade 10 of MIL-L-21260, filled to operating level. Transfer assembly, control differential, and final drives shall contain lubricating oil conforming to type I, Grade 10 or 30, as applicable, of MIL-L-21260, filled to operating level. DD Form 1397 shall be annotated with type and grade of lubricant used (see 3.1.20).

3.1.8 Engine crankcases. Crankcases of compression or spark ignition engines shall be filled to operating level with lubricating oil conforming to type I of MIL-L-21260 of the seasonal grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with type and grade of lubricant used (see 3.1.20).

3.1.9 Engine preservation. Engine preservation shall be in accordance with 3.1.9.1 through 3.1.9.2 without interruption, except for the cooling specified in 3.1.11.

3.1.9.1 Fuel system and combustion chamber. Prior to processing engine, the engine shall be cooled to assure that cylinder head temperature, measured at the injector nozzle flange surface of all cylinders, is not more than 100°F. Cooling shall be accomplished by induced air currents, circulation of engine coolant, or by waiting the period of time required to arrive at the specified temperature. When ambient temperature exceeds 100°F, engine shall be cooled to a temperature equivalent to the ambient. After engine has been cooled, the fuel supply system from the fuel tank shall be shut off. A portable container with two compartments shall be positioned to provide gravity feed to the engine. One compartment shall be filled with lubricating oil conforming to type I of MIL-L-6002 to which has been added an oil soluble red dye conforming to MIL-D-81298, in a concentration sufficient to impart a marked coloring to the oil. The second compartment shall be filled with diesel fuel conforming to VV-F-800. Connect fuel line from

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auxiliary fuel container to fuel pump supply line. Disconnect lines to secondary fuel filter and connect to slave filter assembly. Disconnect residual vehicle fuel return line at quick disconnect coupling. Connect a transparent plastic fuel line to the engine end of the disconnected fuel return line, insert the other end of the plastic fuel line into a recovery container to collect the residual return oil. Disconnect air cleaner hose from engine air box blower. An air restrictor boot shall be installed on the engine air box blower and secured to assure that air to engine is completely shut off. The fuel valve on the auxiliary fuel tank is completely shut off. The fuel valve on the auxiliary fuel tank shall be turned to the lubricating oil position (grade 1, MIL-L-46002). Place engine fuel control in run position. Crank engine with starter (NOTE: Engine may fire for approximately five (5) seconds) and for a minimum of 30 seconds and not exceeding a maximum of 45 seconds. The fuel valve on the auxiliary fuel container shall be turned to the off position. CAUTION: Special precautions shall be taken to assure that time limits specified shall not be exceeded or the engine starter or starter solenoid may be damaged.

3.1.9.2 Engine purging. Purge fuel return lines of all oil, lubricating, MIL-L-46002, by turning the auxiliary fuel tank valve to the diesel fuel position and close engine fuel control shut-off. Crank engine with starter for a period of time to remove all traces of red colored lubricating oil from plastic return line. (CAUTION: Do not operate starter for a period exceeding 45 seconds.) If preservative oil has not been purged, rest starter for three minutes and repeat purge operation. Turn the auxiliary fuel tank valve to the OFF position. Remove fuel line, and slave filter and reconnect. Turn on vehicle fuel supply system. (Remove air restrictor.) Disconnect plastic fuel return line and reconnect the residual vehicle fuel return at the quick disconnect coupling.

3.1.9.3 Preservation through air intake and exhaust system. After preservation in accordance with 3.1.9.1 through 3.1.9.2, one ounce of preservative oil conforming to grade 1 of MIL-L-46002 shall be atomize sprayed into exhaust opening. Opening shall then be sealed with tape conforming to class IV of PPP-T-60.

The hose shall be disconnected at the air intake, and one ounce of preservative oil conforming to grade 1, MIL-L-46002, shall be atomize sprayed into the air intake opening. The air intake opening and the engine crankcase breathers shall then be sealed with plastic plugs conforming to MIL-C-5501 or with tape conforming to class IV of PPP-T-60. A red warning tag bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR

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PLUGS, AND REINSTALL HOSE" shall be placed in a conspicuous location within driver's compartment. DD Form 1397 shall be annotated to show engine is preserved with VCI and oil (see 3.1.20).

3.1.9.4 Preservation through oil level gage rod opening. After preservation in accordance with 3.1.9.1 through 3.1.9.2, the oil level gage rod shall be removed, and six ounces of preservative oil conforming to grade 1 of MIL-L-46002 shall be atomize sprayed into the crankcase through the oil level gage rod opening. An extension of sufficient length to permit the spray nozzle to be within the crankcase shall be used. Spray nozzle shall not be submerged in the crankcase oil. After spraying has been accomplished, oil level gage rod shall be reinstalled. All openings to the engine interior including oil level gage rod opening and oil filler cap shall be sealed with tape conforming to class IV of PPP-T-60.

3.1.10 Fuel tank. Fuel tank shall be drained to the maximum extent possible. Fuel tank cap and filler screen shall be removed and coated with lubricating oil conforming to type I, grade 30 of MIL-L-21260. One quart of lubricating oil conforming to type I, grade 10 of MIL-L-21260 shall be added to each five gallons or portion thereof of residual fuel that remains after draining tank. Tank cap and filler screen shall be reinstalled.

3.1.11 Cooling system. Prior to, or concurrently with the other preservation, requiring engine operation, cooling system shall be protected by the following procedure. Cooling system shall contain a clean solution consisting of equal parts by volume of antifreeze (ethylene glycol) conforming to type I of O-A-548, and water, filled to capacity. Engine shall be operated until temperature has been reached that causes thermostat to open and assure complete mixing and even distribution of the antifreeze solution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH WATER AND ANTIFREEZE SOLUTION (ETHYLENE GLYCOL) IN EQUAL PARTS BY VOLUME - DO NOT DRAIN," shall be securely attached to radiator filler neck. When tested as specified in 4.6.1, coolant shall show protection to minus 40°F. DD Form 1397 shall be marked to indicate coolant used.

3.1.12 Batteries, cables, and retainers.

3.1.12.1 Battery supports and retainers. Battery supports and retainers shall be preserved with compound conforming to MIL-C-450.

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3.1.12.2 Dry charged batteries and cables. Dry charged batteries shall be installed in the vehicle battery carrier. Filler cap openings shall be sealed by placing a 2-inch wide by 3-mil thick piece of film conforming to type II of MIL-F-22191 over all filler cap openings with caps removed. The film shall be of sufficient length to allow the film to be depressed into the filler cap opening to the same depth as the filler cap. Filler caps shall be screwed into the filler openings to form a complete seal without damaging plastic film. Battery cables shall be secured to battery carrier with 3/4-inch wide tape conforming to Type IV of PPP-T-97. If batteries have been processed in accordance with MIL-B-11188 they need not be reprocessed as above.

3.1.12.3 Electrolyte. Electrolyte shall be packaged, packed, and marked as specified for type IV, class 1 or 2 unit, as applicable, in accordance with MIL-S-207. The packed electrolyte shall be stowed with the basic issue items of on vehicle equipment (OVE) and secured independently to permit separate removal.

3.1.13 Fire control items.

3.1.13.1 Quadrant, periscopes, and telescopes. Quadrant, periscopes, and telescopes shall remain installed. Immediately after cleaning (see 3.1.5.1.3), optics shall be covered with a four layer thickness of lens tissue conforming to NNN-P-40. Lens tissue shall be secured with tape conforming to PPP-T-42. All exposed, unpainted metal surfaces shall be coated with grease conforming to MIL-G-10924.

3.1.13.2 Vehicles with panoramic telescope ballistic cover assembly installed. Remove cover assembly and clean in accordance with process C-1 of MIL-P-116. All unpainted surfaces shall be coated with preservative conforming to grade 4 of MIL-C-16173. Cover assembly shall be packaged and packed, placed inside Crew Compartment, and blocked and braced to prevent movement during shipment. Close opening with plywood cover 17" diameter x 1/2" thick, secure with 3 bolts (3/8" x 1 1/2") with nut and washer.

3.1.14 Cannon and mount.

3.1.14.1 Cannon. Immediately after cleaning (see 3.1.5.2.1), bore and chamber of the cannon shall be coated with preservative oil conforming to VV-L-800. Excess preservative shall be allowed to drain from coated surfaces. A strip of volatile corrosion inhibitor (VCI) treated carrier material conforming to type I, class 3, style A of MIL-P-3420 (see 3.1.3.1) shall be cut and rolled into a tube with the

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VCI treated surface on the outside. The carrier material shall be of a size that will provide a continuous cover for the bore and the chamber surface. The rolled carrier material shall be inserted into cannon, extending entire length of bore and chamber. The carrier tube shall not be forced or kinked in a manner that will obstruct the chamber. The cannon shall remain in battery with the cannon tube secured in travel lock position.

3.1.4.1.1 Muzzle plug. A plug shall be provided for the muzzle end of cannon. Muzzle plug shall be completely overwrapped with aluminum foil conforming to QQ-A-1876, positioned in muzzle end of cannon and secured in place with tape conforming to class 1 of PPP-T-60. The joint around muzzle plug and cannon shall be completely sealed with tape conforming to class 1 of PPP-T-60. A polyolefin bag conforming to type I, class 1, grade A, finish 1 of L-P-378 shall be provided. Bag shall be 16 inches long, of applicable width and 0.006 inch thick. Bag shall be installed over muzzle end of cannon and secured in place with four strips of tape conforming to class 1 of PPP-T-60. Tape shall run lengthwise on the cannon placed at top, bottom, and each side of cannon. Tape shall be 12 inches long and minimum 1 inch in width, applied equally 6 inches onto bag and 6 inches onto painted surfaces of the cannon. Bag shall be sealed to cannon with tape conforming to class 1 of PPP-T-60. Tape shall be of applicable length to provide a continuous seal around circumference of cannon, and 6 inches in width. Tape shall be applied equally 3 inches onto bag and 3 inches onto painted surfaces of the cannon. Two additional 1 inch strips of tape conforming to class 1 of PPP-T-60 shall be applied at equal intervals between muzzle end of cannon and area where bag is sealed to cannon. Tape shall be applied completely around circumference of bag to provide additional securement of bag to cannon.

3.1.14.1.2 Breech mechanism. All unpainted surfaces, including phosphated surfaces of the breech block, breech operating mechanism and firing mechanism, shall be coated with grease conforming to MIL-G-10924. Breech block shall be closed.

3.1.14.2 Muzzle brake. Muzzle brake (M109) shall be removed and cleaned in accordance with process C-1 of MIL-P-116. All unpainted surfaces and threaded areas of muzzle brake shall be coated with preservative conforming to grade 4 of MIL-C-16173. Muzzle brake shall be placed inside crew compartment and blocked and braced to prevent movement during shipment.

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3.1.14.3 Exercising of recoil mechanism. Prior to processing as specified in 3.1.14.3.1 and when the recoil mechanism has not been exercised, proof fired, overhauled, or manufactured within 4 months prior to preparation for storage or shipment, the recoil mechanism shall be exercised a minimum of three extensions of the recoil piston. Extension shall be a minimum of 6 inches. Record of exercising shall be entered on DA Form 2408-4 "Weapon Record Data" and proof testing of the weapon shall be entered on DA Form 2408-9 "Proof Acceptance Record" (see 3.1.20).

3.1.14.3.1 Processing of recoil mechanism after exercising. Accessible machined surfaces of the cannon immediately forward of the recoil mechanism shall be coated with grease conforming to MIL-G-10924. Inaccessible machined metal surfaces shall be fogged with preservative oil conforming to grade 1 of MIL-L-21260. The surface of the recoil mechanism immediately forward of breech ring collar shall be coated with grease conforming to MIL-G-10924 and overwrapped with barrier material conforming to grade C of MIL-B-121 and sealed with tape conforming to class 1 of PPP-T-60.

3.1.14.3.1.1 Processing bore evacuator. The front and rear joints of the bore evacuator to the gun tube shall be sealed with tape conforming to type I, PPP-T-60.

3.1.14.3.2 Counter recoil mechanism. Without disassembly, exposed unpainted mechanism surfaces of the counter recoil mechanism shall be cleaned, then coated with grease conforming to MIL-G-10924.

3.1.14.4 Exercising of replenisher. Replenisher assemblies shall be exercised coincidentally with the recoil mechanism (see 3.1.14.3).

3.1.14.4.1 Processing of replenisher after exercising. Replenisher shall be filled to bleed position with hydraulic fluid conforming to MIL-H-6083, then drained to operating level.

3.1.14.5 Elevating mechanism. Exposed unpainted surfaces of elevating pinion and rack, and elevating hand crank shaft shall be coated with grease conforming to MIL-G-10924.

3.1.14.6 Traversing mechanism. Exposed unpainted surfaces of traversing gear pinion, and traversing hand crank shaft shall be coated with grease conforming to MIL-G-10924. Cannon mount travel lock and cab lock of the M109, M109A1, M109A1B and M109A2 vehicles shall be secured.

3.1.14.7 Equilibrators. Equilibrators shall be disassembled. The equilibrator chain, sheave, and all unpainted surfaces shall be coated with grease conforming to MIL-G-10924.

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3.1.14.8 Rammer for M109, M109A1, M109A1B, and M109A2. Exposed unpainted surfaces of rammer cylinder piston rod and assembly, (including the wire counterbalance, where applicable), shall be coated with grease conforming to MIL-G-10924.

3.1.14.9 Hydraulic systems. Hydraulic systems shall be filled to operating level with hydraulic fluid conforming to MIL-H-6083.

3.1.15 Spade for M109, M109A1, M109A1B, and M109A2. The ends of hinge pins of the spade shall be coated with grease conforming to MIL-G-10924.

3.1.16 Backrest and seats. Immediately after drying (see 3.1.5.1.2) backrests and cushioned components of seats shall be covered with paper having a minimum basis weight of 60 pounds, conforming to grade B of UU-P-268. Paper shall be secured in place with tape conforming to PPP-T-42.

3.1.17 Hatches and doors. Rubber seats around hatches and doors shall be coated with powdered talc conforming to type IV, class C of ZZ-T-416, or talc, technical T1, MIL-T-50036. Hatches and doors shall be closed and locked from the inside except that for shipment, commander's hatch and rear door shall be secured using an approved railroad type seal placed through locking ring.

3.1.18 Miscellaneous preservation. Unless otherwise specified, all exposed unpainted machined-metal surfaces on the interior of the vehicle shall be coated with preservative conforming to grade 4 of MIL-C-16173. All exposed, unpainted machined-metal surfaces on the exterior, except the track shoes, shall be coated with preservative conforming to grade 1 of MIL-C-16173.

3.1.19 On vehicle equipment (OVE). Unless otherwise specified (see 6.1), all spare parts, tools and equipment shall be preserved, packaged and packed in accordance with requirements of MIL-P-12841. Pack shall be identified to pertinent vehicle by serial number and, except during shipment, shall be stored inside buildings.

3.1.20 Record forms. A copy of "The Equipment Log Book" consisting of Binder (FSN 7510-889-3494) with required forms, and two copies of "Processing and Deprocessing Record for Shipment, Storage and Issue of Vehicles and Spare Engines," DD Form 1397, and single copies of DA Form 2408-4 Weapon Record Data, and DA Form 2408.9 Proof Acceptance Record (see 6.2) shall be furnished and completed with each vehicle. Instructions for preparation, attachment and location of "The Equipment Log Book" shall be supplied by the Government representative. One copy

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of DD Form 1397 shall be placed in a bag conforming to type II, class B of MIL-B-117. Bag shall be closed by stapling, identified, and secured to headlamp. The duplicate copy shall be placed with the log book.

3.1.21 Vehicle closure. Each vehicle shall be provided with a vehicle closure kit (commonly called covered wagons). The closure kit shall be fabricated, assembled and installed in accordance with their respective drawing; M108, M109, M109A1 vehicle closure drawing 10941519; M109A1B vehicle closure kit drawing 12260206. All sharp corners of framework and bows where cloth cover will make contact, such as corners on base of the frame, shall be cushioned with 3/4 inch minimum thickness of cushioning material conforming to PPP-C-843, secured in place with tape conforming to class 1 of PPP-T-60. To lift vehicles for loading, cover shall be rolled away from front and rear bows to expose vehicle lifting eyes.

3.1.21.1 Ventilation. Driver's compartment access cover, and transmission access covers and gaskets from the underside of vehicle shall be removed for ventilation. Unpainted metal surfaces, and surfaces exposed by removal of these items shall be coated with preservative conforming to grade 1 of MIL-C-16173. Covers and gaskets shall be placed in a box conforming to PPP-B-636, identified, and stowed within vehicle in a manner preventing movement in transit. The following information shall be stenciled on the exterior of the vehicle in a conspicuous location: "REMOVE SCREENS, INSTALL ACCESS COVERS AND GASKETS BEFORE OPERATION." Stenciling color shall be white or yellow, using paint conforming to MIL-P-13983. Stenciling shall be in characters 3/4 inch high minimum.

3.1.21.1.1 Screens. Installation of screens shall be in accordance with ventilation kit drawing 12260223.

3.1.22 Fire extinguishers. Fire extinguishers shall have a minimum of 90 percent of the rated full charge. All seats shall be intact. All exterior emergency handles shall be covered with tape. DA Form 253 shall be completely and securely attached to each extinguisher (see 6.2).

3.1.23 OVE containers.

3.1.23.1 Domestic shipment. Unless otherwise specified (see 6.1) for shipping purposes, OVE containers shall be placed on floor of railroad cars and blocked, braced, and anchored to prevent movement in transit. Corner protectors shall be used under strapping.

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3.1.23.2 Oversea shipment. OVE containers shall be placed on the OVE saddle in a position which will not increase the overall cube of the vehicle. Each large wood box such as manufacturer's or depot OVE shall be secured with 2x4 blocking, fore and aft, and with two (2) 1 1/4 inch strapping crosswise of box. Smaller OVE boxes shall be secured to larger OVE boxes, or to OVE saddle with 3/4 inch strapping, or secured in interior of cab. 1 1/4 and 3/4 inch strapping shall conform to type I, class B of QQ-S-781. Corner protectors shall be used under strappings.

3.1.24 OVE saddle. Vehicle shall be provided with an OVE saddle constructed, assembled and installed in accordance with drawing 12260160.

3.1.25 Tow hooks. Tow hooks and related hardware, except pintle, shall be removed for shipment and packaged in a type CF, class weather resistant box conforming to PPP-B-636. Box shall be closed with tape conforming to PPP-T-42, identified as to contents, and securely stowed within personnel compartment.

3.1.26 Closure markings. The information "TO LIFT VEHICLE FOR LOADING, UNFASTEN AND ROLL COVER FROM FRONT AND REAR BOWS EXPOSING LIFTING EYES - AFTER LOADING, SECURE COVER IN THE ORIGINAL POSITION" shall be stenciled on the front and rear of the closure in characters a minimum of 3/4 inch high, using white enamel conforming to TT-E-529.

3.1.26.1 Closure disposition marking. The following information shall be stenciled on the outside, front and rear of the cover: "REUSABLE CLOSURE (COVER AND FRAMEWORK) - DO NOT DESTROY - WHEN REMOVED AND NO LONGER REQUIRED FOR PROTECTION, PACKAGE AND SHIP PER INSTRUCTIONS ON SIDE OF COVER." The following information shall be stenciled on the inside, front and rear of the cover:

PACKAGING AND SHIPPING INSTRUCTIONS:

1. DEPOT OVE CONTAINER MAY BE USED AS SHIPPING BOX.
2. SECURELY BUNDLE LIKE ITEMS.
3. PACKAGE HARDWARE IN CLOTH BAGS.
4. GROUP LARGEST, HEAVIEST ITEMS IN BOTTOM OF BOX.
5. PLACE OTHER ITEMS IN VOID BETWEEN ABOVE ITEMS.
6. FOLD COVER - PLACE ON TOP OF ABOVE ITEMS.
7. IMMOBILIZE ALL ITEMS AND SECURE COVER.
8. SHIP TO (address to be furnished by contracting officer).

All stenciled characters to be 3/4 inch high minimum height, in white enamel conforming to TT-E-529.

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3.1.26.2 Depot OVE box marking. The information "DO NOT DESTROY - USE FOR RETURN SHIPMENT OF VEHICLE CLOSURE KIT" shall be stenciled on the depot OVE box in lettering a minimum of 3/4 inches high with a contrasting color of enamel conforming to TT-E-529.

3.2 Level B. Vehicles shall be processed in the same manner as specified for level A, with the following exceptions:

3.2.1 Transmission and final drive. Transmission and final drives shall contain normal operational lubricant as specified on lubrication order, filled to operating level. DD Form 1397 shall be annotated to indicate grade of lubricant used.

3.2.2 Engine crankcase. Engine crankcase shall contain normal operational lubricant as specified on lubrication order, filled to operational level. DD Form 1397 shall be annotated to indicate grade of lubrication used.

3.2.3 Engine preservation. The engine shall require no preservation for level B shipment and storage.

3.2.4 Air cleaners. The air cleaners shall neither be disassembled nor sealed for level B shipment and storage.

3.2.5 Fuel tanks. Unless otherwise specified (see 6.1) vehicles shall be shipped without draining residual fuel from the fuel tanks.

3.2.6 Personnel heater and fuel pump. Unless otherwise specified (see 6.1), personnel heaters and fuel pumps shall be in a ready-to-use condition.

3.2.7 Ventilation. All access covers and gaskets on the underside of the vehicle shall be removed, packaged and stowed as specified in paragraph 3.1.21.1. The following information shall be stenciled on the exterior of the vehicle in a conspicuous location: "REINSTALL ACCESS COVERS AND GASKETS ON UNDERSIDE OF VEHICLE BEFORE OPERATING VEHICLE."

3.2.8 Vehicle closure. Vehicle closure kit shall not be provided for level B shipment and storage.

3.3 All levels.

3.3.1 Loading. Loading of vehicles on open top railroad cars shall be in accordance with the applicable requirements of section 1 of the Association of American Railroads Publication, General Rules (see 2.2).

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3.3.1.1 Loading vehicles on special rub rail cars. Vehicles shall be loaded and secured on special rub rail cars in accordance with section 6, figure 87 of the Association of American Railroads Publication (see 2.2). Transmission shall be placed in neutral position. Brake levers shall be secured in released position to ensure lateral movement of vehicles on car in transit.

3.3.2 Reprocessing engine after loading, level A. If engine is operated in connection with moving of vehicle to loading area, or during the loading or unloading itself, the engine shall be reprocessed as specified in 3.1.9.1 through 3.1.9.2. Vehicle cover shall be rolled clear of engine intake and exhaust to provide air circulation and to prevent damage to cover. After reprocessing engine, vehicle closure shall be replaced in its original position.

3.3.2.1 Level B. If engine is operated in connection with movement of vehicle for loading or unloading, there shall be no additional processing of engine.

3.3.3 Drive on - drive off capability. When vehicles are to be operated for loading or unloading, the following provisions apply:

3.3.3.1 Fuel tanks. As specified (see 6.1), additional fuel shall be added, as required, to accomplish movement of vehicle.

3.3.3.2 Batteries and electrolyte. New batteries with electrolyte added and fully charged will be installed in the vehicle and battery cables properly connected.

3.4 Marking. In addition to any special marking required herein or in the contract or order (see 6.1), vehicles shall be marked in accordance with MIL-STD-129. The information "LIFT HERE" shall be stenciled adjacent to each lifting eye, using white enamel conforming to TT-E-529. Stenciling shall be in characters a minimum of 3/4 inches high and shall be readily visible when OVE is loaded on OVE rack.

3.4.1 Closure marking. The information "TO LIFT VEHICLE FOR LOADING, UNFASTEN AND ROLL COVER FROM FRONT AND REAR BOWS EXPOSING LIFTING EYES," shall be stenciled in a conspicuous location on the front and rear of vehicle closure, using white enamel conforming to TT-E-529. Stenciling shall be in characters a minimum of 3/4 inches high. The information "CLOSURE KIT, VEHICLE PROTECTIVE - (M108, M109, M109A1, M109A1B, M109A2 as applicable) - DO NOT DESTROY - REINSTALL FOR SHIPMENT OR STORAGE OR SHIP WITH OVE," shall be stenciled in a conspicuous location on one end and one side of vehicle closure, using white enamel conforming to TT-E-529. Stenciling shall be in characters 2 inches high minimum.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First production processed and one of the next 10 production processed vehicles. The first production processed vehicle (see 3.1.1), and one of the next 10 production processed vehicles (see 3.1.2) shall be subjected to the inspection and tests specified in 4.5 through 4.6.1.

4.3 Production processed vehicles. Unless otherwise specified, all production processed vehicles shall be subjected to the inspection specified in 4.5.2 through 4.5.2.3.

4.4 Rejection. Failure of the first production, one of the next 10, or of any production processed vehicle, to conform to the requirements of this specification shall be cause for rejection of the vehicle, and refusal to continue acceptance of vehicles by the Government until objective evidence has been provided by the contractor that corrective action has been taken to eliminate the condition which caused rejection.

4.5 Quality conformance inspection.

4.5.1 Materials. Except for materials which have been Government inspected at the source, all materials to be used in processing of vehicles shall be inspected in accordance with the material specifications; or certified inspection and laboratory test reports shall be furnished which show that material, as furnished, conforms to the detailed specification.

4.5.2 Processing. All vehicle processing shall be inspected to determine conformance to this specification. Inspection of processing shall include all items specified in table 1, and 4.5.2.1 through 4.6.1.

4.5.2.1 Cleaning. To determine conformance to 3.1.5.1, interior of vehicles shall be examined for cleanliness, and one vehicle each day shall be tested for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.1.5.2, exterior

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of vehicle shall be examined for cleanliness. Surfaces to which tape is to be applied shall be examined for cleanliness before applying tape.

4.5.2.2 Fuel tank. To determine conformance to 3.1.10 interior of fuel tank shall be examined and visual inspection shall assure that complete processing, as specified, has been accomplished.

TABLE I. Processing inspection.

Component	Cleaning Levels A & B	Preservation Level A	Preservation Level B	Packaging Levels A & B
Preparation prior to processing				3.1.4
Processing records				3.1.4.1
Disassembly		3.1.4.2	3.1.4.2	3.1.4.2
Match marking		3.1.4.2.1	3.1.4.2.1	3.1.4.2.1
Interior of vehicles	3.1.5.1			
Battery supports and retainers	3.1.5.1.1	3.1.5.1.1	3.1.5.1.1	
Backrests and seats	3.1.5.1.2	3.1.5.1.2	3.1.5.1.2	3.1.16
Fire control items	3.1.5.1.3	3.1.13	3.1.13	3.1.13
Exterior of vehicle	3.1.5.2			
Cannon	3.1.5.2.1	3.1.14	3.1.14	
Transmission and final drive 1/		3.1.7	3.2.1	
Engine crankcase		3.1.8	3.2.2	
Engine preservation		3.1.9	3.2.3	
Fuel tank		3.1.10	3.2.5 and 3.3.3.1	
Cooling system 1/		3.1.11	3.1.11	
Batteries and cables		3.1.12.1	3.3.3.2	3.1.12.2
Electrolyte				3.1.12.3
Spade for M109 series		3.1.15		
Hatches and doors				3.1.17
Miscellaneous preservation		3.1.18	3.1.18	
On vehicle equipment		3.1.19	3.1.19	3.1.19
Record forms 2/				3.1.20
Vehicle closure		3.1.21	3.2.8	3.1.21
Ventilation		3.1.21.1	3.2.7	
Fire extinguisher				3.1.22
OVE containers				3.1.23
OVE saddle				3.1.24
Tow hooks				3.1.25

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TABLE I. Processing inspection. - Continued

Component	Cleaning Levels A & B	Preservation Level A	Preservation Level B	Packaging Levels A & B
Loading				3.3
Reprocessing engine after loading		3.3.2	3.3.2.1	
Marking				3.4
Drive on - drive off capability			3.5	

1/ Inspect DD 1397.
2/ Inspect DA Form 2408-4 and DA Form 2408-9.

4.5.2.3 Cannon. Cannon shall be examined to determine condition and effectiveness of processing. When reprocessing has been accomplished, cannon shall be examined for conformance to 3.1.5.2.1 and 3.1.14.

4.5.2.4 Engine. To determine conformance to 3.1.9, interior of engine from the first production processed vehicle, and one of the next 10 production processed vehicles shall be examined for surface coverage. Engine shall be disassembled to extent necessary to permit visual examination of all surfaces within the combustion chamber. (NOTE: The combustion chamber shall be considered as all surfaces within the cylinder, from and including the crown of the piston, to and including the surface of the head within the cylinder.) All surfaces within the combustion chamber shall have a "wet" coating of the preservative oil such as obtained when the item is dipped or flushed with the preservative oil. The processing method of the approved preserved engine shall be submitted in writing for production on all subsequent engines.

4.6 Tests.

4.6.1 Cooling system. To determine conformance to 3.1.11.1, one vehicle from each day's production shall be selected at random and the engine coolant tested using an antifreeze tester conforming to GG-T-250, with a range of minus 60° to plus 160°F.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following;

(a) Title, number, and date of this specification.
(b) Type of processing required (see 1.2).
(c) If inspection of the first production processed vehicle is other than specified (see 3.1.1).
(d) If packaging and packing of OVE is other than specified (see 3.1.19).
(e) If domestic shipment of OVE containers is other than specified (see 3.1.23.1).
(f) If draining of residual fuel from tanks is required (see 3.2.5).
(g) Special marking required (see 3.3.3).

6.2 Forms. A copy of "The Equipment Log Book" and all required forms will be furnished the contractor by the Government at least 30 days before shipment of the equipment as required by the contract delivery schedule (see 3.1.20).

6.3 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Army - AT

User activity:
Army - MT

Review activities:
Army - SM, AR

Preparing activity:
Army - AT

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